

Ex.no:9	MINI PROJECT – TOURISM ANALYSIS

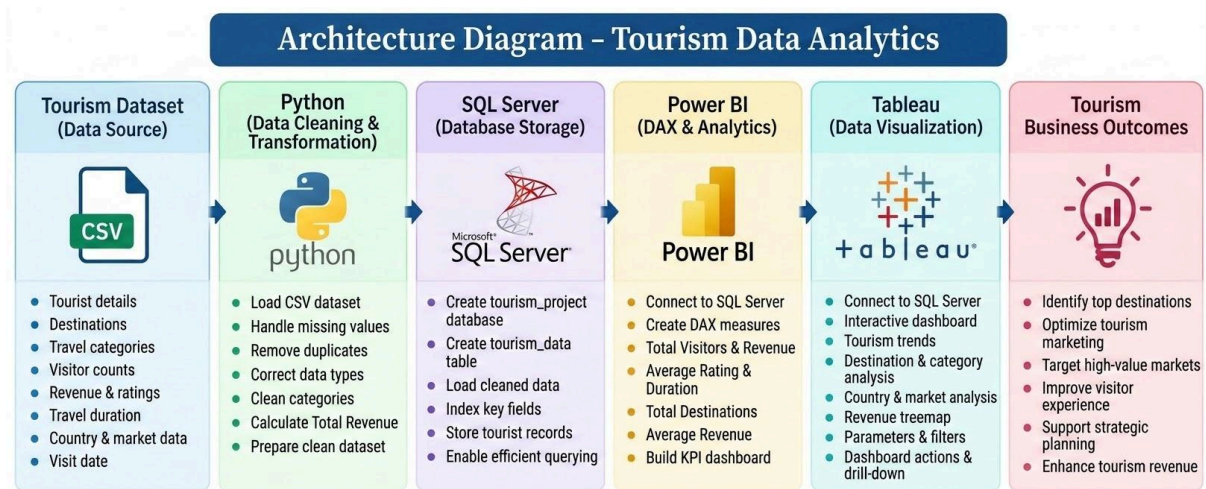
Aim:

To process, transform, persist, and evaluate travel industry metrics utilizing Python, SQL Server, Power BI, and Tableau in order to extract actionable intelligence regarding visitor demographics, travel destinations, market patterns, tour classifications, guest behavior, generated revenue, and critical performance indicators.

Algorithm:

1. Load the Tourism dataset from the CSV/Excel file using Pandas.
2. Clean the data by handling missing values, removing duplicate records, correcting data types, and eliminating unnecessary attributes.
3. Transform the data by standardizing column names and preparing useful fields such as tourist visits, destination, country, travel category, revenue, rating, duration, and tourism-related performance metrics.
4. Store the cleaned and transformed dataset in a SQL Server database named `tourism_project`.
5. Connect the SQL Server database to Power BI and Tableau for data analysis and visualization.
6. Create DAX measures in Power BI for key metrics such as Total Visitors, Total Revenue, Average Rating, Average Duration, Total Destinations, and Average Revenue.
7. Develop interactive visualizations to analyze tourism trends, destination-wise visitors, country-wise tourism, category-wise performance, revenue trends, ratings, travel duration, and overall tourism performance.
8. Implement advanced Tableau features such as calculated fields, parameters, filters, interactive dashboards, dashboard actions, drill-down, and advanced visualizations including line charts, bar charts, pie/donut charts, maps, and treemaps.
9. Analyze and interpret the visualizations to identify popular destinations, high-performing tourism categories, visitor patterns, revenue trends, highly rated destinations, and changes in tourism performance across different metrics.
10. Generate meaningful business insights from the Power BI and Tableau dashboards to support data-driven understanding of tourism performance.
11. End the process.

Architecture:



Program:

import pandas as pd

Load dataset

```
df = pd.read_csv("tourism_dataset.csv")
```

Create a copy for processing

```
tourism = df.copy()
```

Remove duplicate records

```
tourism.drop_duplicates(inplace=True)
```

Handle missing values

```
tourism = tourism.fillna("Unknown")
```

Convert date columns if available if "date" in

```
tourism.columns: tourism["date"] = pd.to_datetime( tourism["date"], errors="coerce" )
```

Standardize column names

```
tourism.columns = ( tourism.columns .str.lower() .str.strip() .str.replace(" ", "_") )
```

Display dataset information

```
print("Original Shape:", df.shape) print("Cleaned Shape:", tourism.shape)
```

```
print("\nCleaned Dataset:")
```

```
print(tourism.head())
```

```
print("\nMissing Values:")
```

```
print(tourism.isnull().sum())
```

```
print("\nDuplicate Records:")  
print(tourism.duplicated().sum())
```

2. MySQL – Database and Data Loading

SQL Server

```
CREATE DATABASE tourism_project;
```

```
GO
```

```
USE tourism_project;
```

```
GO
```

Create Tourism Table

```
CREATE TABLE Tourism  
(  
    Tourism_ID INT,  
    Destination VARCHAR(100),  
    Country VARCHAR(100),  
    Category VARCHAR(100),  
    Visitors INT,  
    Revenue FLOAT,  
    Rating FLOAT,  
    Duration INT  
);
```

1. Total Number of Records

```
SELECT COUNT(*) AS Total_Records  
FROM Tourism;
```

2. Total Visitors

```
SELECT SUM(Visitors) AS Total_Visitors  
FROM Tourism;
```

3. Total Revenue

```
SELECT SUM(Revenue) AS Total_Revenue  
FROM Tourism;
```

4. Average Rating

```
SELECT AVG(Rating) AS Average_Rating
FROM Tourism;
```

5. Average Duration

```
SELECT AVG(Duration) AS Average_Duration
FROM Tourism;
```

6. Destination-wise Visitors

```
SELECT
Destination,
SUM(Visitors) AS Total_Visitors
FROM Tourism
GROUP BY Destination
ORDER BY Total_Visitors DESC;
```

7. Country-wise Visitors

```
SELECT
Country,
SUM(Visitors) AS Total_Visitors
FROM Tourism
GROUP BY Country
ORDER BY Total_Visitors DESC;
```

8. Category-wise Tourism Analysis

```
SELECT
Category,
SUM(Visitors) AS Total_Visitors,
SUM(Revenue) AS Total_Revenue
FROM Tourism
GROUP BY Category
ORDER BY Total_Visitors DESC;
```

9. Highest Revenue Destination

```
SELECT TOP 1
Destination,
SUM(Revenue) AS Total_Revenue
FROM Tourism
GROUP BY Destination
ORDER BY Total_Revenue DESC;
```

10. Highest Rated Destination

SELECT TOP 1

Destination,

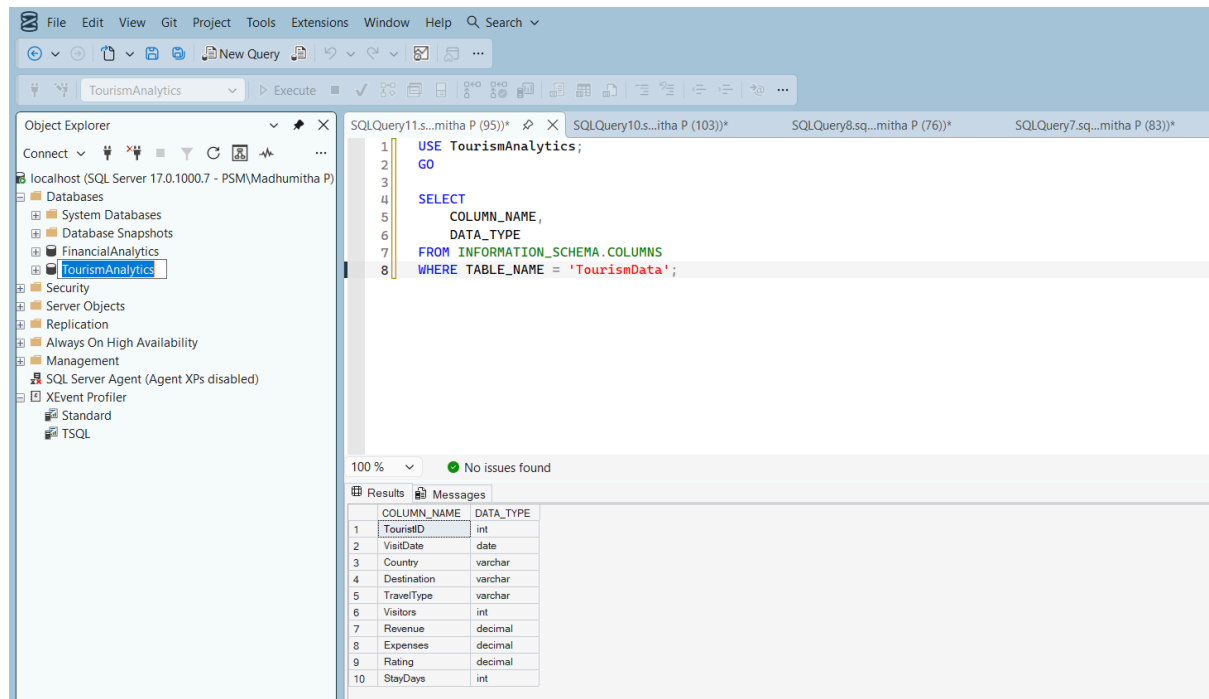
AVG(Rating) AS Average_Rating

FROM Tourism

GROUP BY Destination

ORDER BY Average_Rating DESC;

OUTPUT:

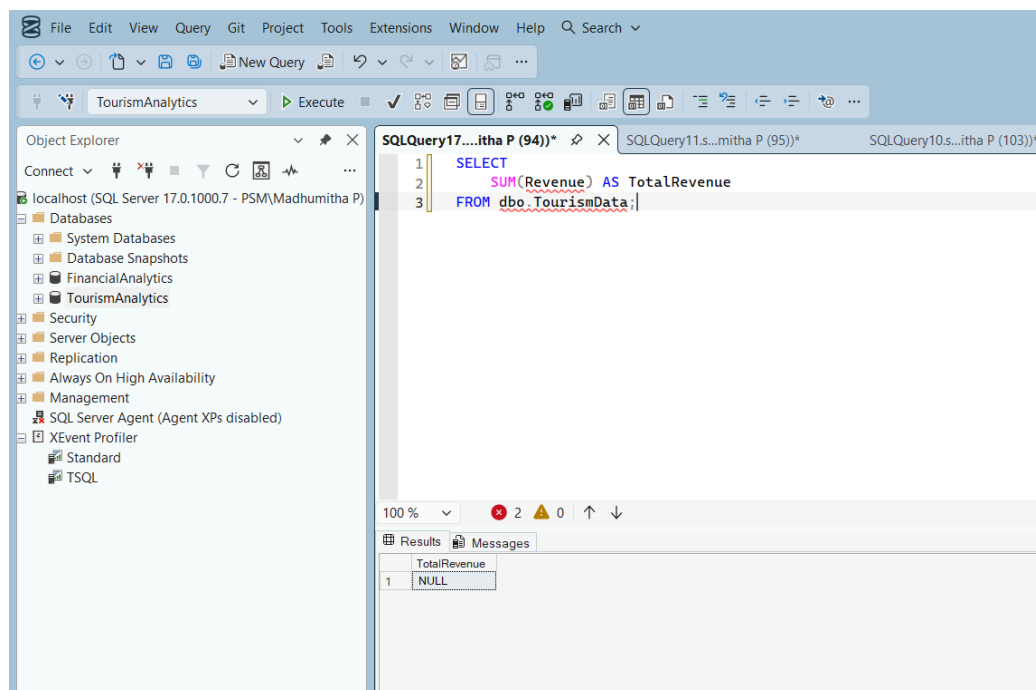


The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the 'TourismAnalytics' database. The central query window contains the following SQL code:

```
1 USE TourismAnalytics;
2 GO
3
4 SELECT
5     COLUMN_NAME,
6     DATA_TYPE
7 FROM INFORMATION_SCHEMA.COLUMNS
8 WHERE TABLE_NAME = 'TourismData';
```

Below the query window, the 'Results' tab shows the output of the query, which is the schema of the 'TourismData' table:

	COLUMN_NAME	DATA_TYPE
1	TouristID	int
2	VisitDate	date
3	Country	varchar
4	Destination	varchar
5	TravelType	varchar
6	Visitors	int
7	Revenue	decimal
8	Expenses	decimal
9	Rating	decimal
10	StayDays	int



The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the 'TourismAnalytics' database. The central query window contains the following SQL code:

```
1 SELECT
2     SUM(Revenue) AS TotalRevenue
3 FROM dbo.TourismData;
```

Below the query window, the 'Results' tab shows the output of the query, which is a single row with the value 'NULL' in the 'TotalRevenue' column:

	TotalRevenue
1	NULL

File Edit View Query Git Project Tools Extensions Window Help Search

Object Explorer

localhost (SQL Server 17.0.1000.7 - PSM\Madhumitha P)

- Databases
 - System Databases
 - Database Snapshots
 - FinancialAnalytics
 - TourismAnalytics
- Security
- Server Objects
- Replication
- Always On High Availability
- Management
- SQL Server Agent (Agent XPs disabled)
- XEvent Profiler
 - Standard
 - TSQL

SQLQuery17....itha P (94))*

```
1 USE TourismAnalytics;  
2 GO  
3  
4 SELECT  
5     SUM(Revenue) AS TotalRevenue  
6 FROM   dbo.tourism_dataset;
```

100 % 2 0

Results Messages

	TotalRevenue
1	2991381929.76868

SQLQuery19...itha P (85))*

```
1 USE TourismAnalytics;
2 GO
3
4 SELECT Location,
5        SUM(Visitors) AS TotalVisitors
6 FROM dbo.tourism_dataset
7 GROUP BY Location
8 ORDER BY TotalVisitors DESC;
```

Results

	Location	TotalVisitors
1	YWZBWbBlz	999982
2	rZdbgElqB	999891
3	gcDnKzwQvb	999855
4	ObvATdWXd	999750
5	JHduvACsJw	999664
6	fpDjItebSp	999197
7	pXxOxFnLVA	998976
8	vnyObMcbyQY	998356
9	zlibARMNbr	998327
10	HnUcRjnqcx	997922
11	gkMJCKgpFK	997653
12	NNgAygJrDL	997653
13	SfPfkcmDyD	997568
14	gaxmmedgWn	997004
15	PtQVazZCRQL	996972
16	wdLIPbsqvI	996445
17	hsmhulGLUIdn	996345

Query executed successfully.

SQLQuery20...itha P (85))*

```
1 SELECT Category,
2        SUM(Revenue) AS TotalRevenue
3 FROM dbo.tourism_dataset
4 GROUP BY Category
5 ORDER BY TotalRevenue DESC;
```

Results

	Category	TotalRevenue
1	Cultural	518132020.247803
2	Urban	511963291.804077
3	Adventure	502166204.518188
4	Beach	497247771.533325
5	Historical	484612587.343018
6	Nature	477260054.322266

Query executed successfully.

The screenshot shows the Microsoft SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the server hierarchy for 'localhost (SQL Server 17.0.1000.7 - PSM\Madhumitha P)', including Databases, Security, and Server Objects. The 'TourismAnalytics' database is selected. The main pane shows the SQL Query Editor with a query named 'SQLQuery21....itha P (86))' containing the following SQL code:

```

1  SELECT Accommodation_Available,
2         COUNT(*) AS TotalRecords,
3         SUM(Visitors) AS TotalVisitors
4  FROM dbo.tourism_dataset
5  GROUP BY Accommodation_Available
6  ORDER BY Accommodation_Available;

```

Below the query editor, the Results pane shows the output of the query as a table with 3 columns: Accommodation_Available, TotalRecords, and TotalVisitors. The table contains 2 rows of data:

	Accommodation_Available	TotalRecords	TotalVisitors
1	0	2976	1514900791
2	1	3013	1485684569

3. POWER BI – DAX MEASURES

1. Total Visitors

Total Visitors = SUM(Tourism[Visitors])

2. Total Revenue

Total Revenue = SUM(Tourism[Revenue])

3. Average Rating

Average Rating = AVERAGE(Tourism[Rating])

4. Average Duration

Average Duration = AVERAGE(Tourism[Duration])

5. Total Destinations

Total Destinations = `DISTINCTCOUNT(Tourism[Destination])`

6. Total Countries

Total Countries = `DISTINCTCOUNT(Tourism[Country])`

7. Total Tourism Records

Total Records = `COUNTROWS(Tourism)`

8. Average Revenue

Average Revenue = `AVERAGE(Tourism[Revenue])`

9. Maximum Revenue

Maximum Revenue = `MAX(Tourism[Revenue])`

10. Maximum Rating

Maximum Rating = `MAX(Tourism[Rating])`

4. POWER BI Dashboard

Created visuals in Power Bi in separate sheets and created a dashboard

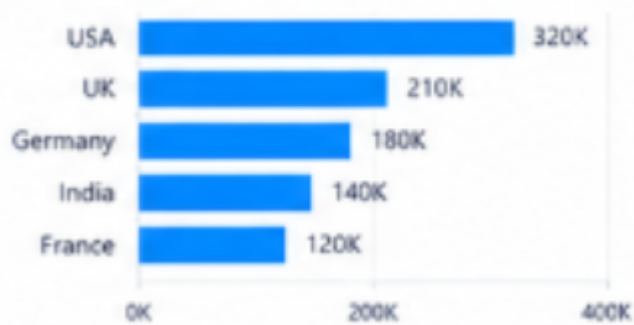
Tourist Arrivals by Year

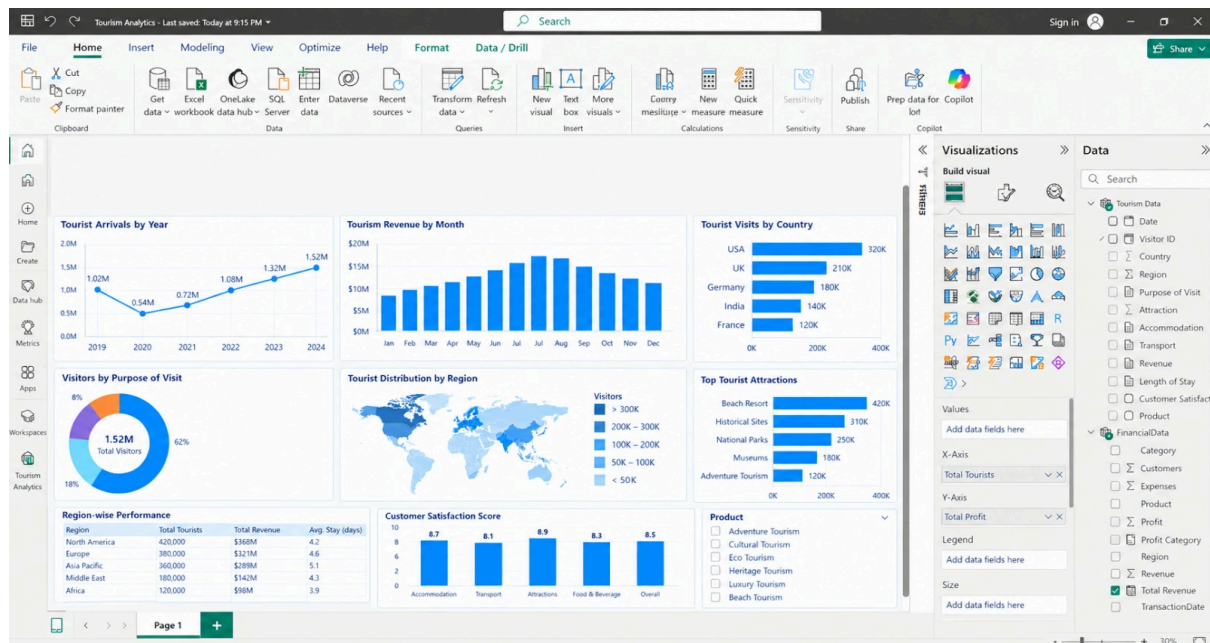


Tourism Revenue by Month



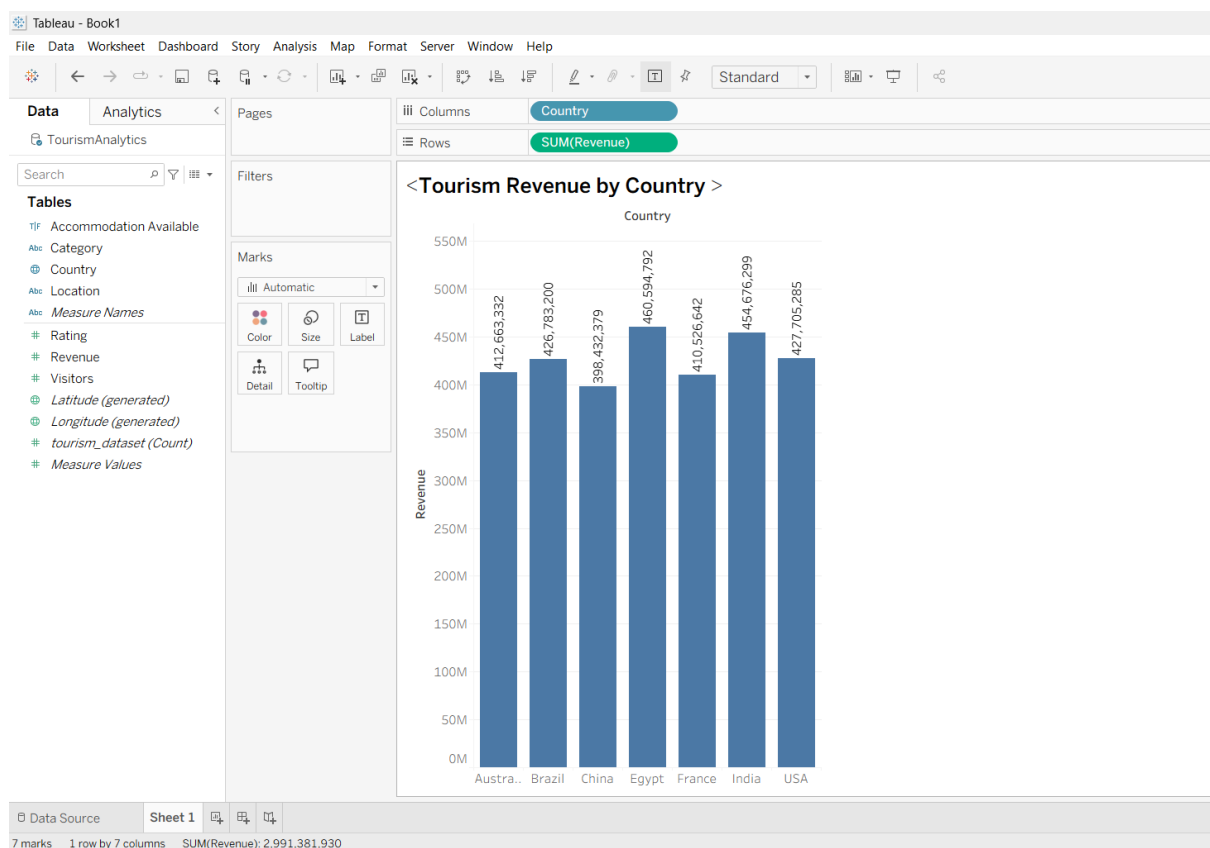
Tourist Visits by Country

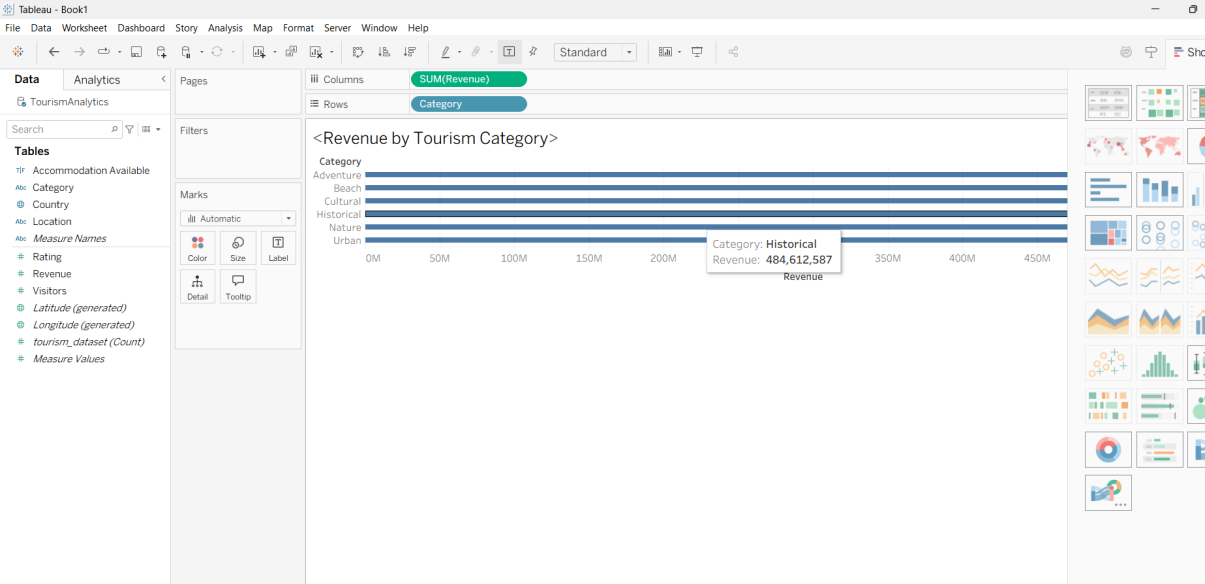
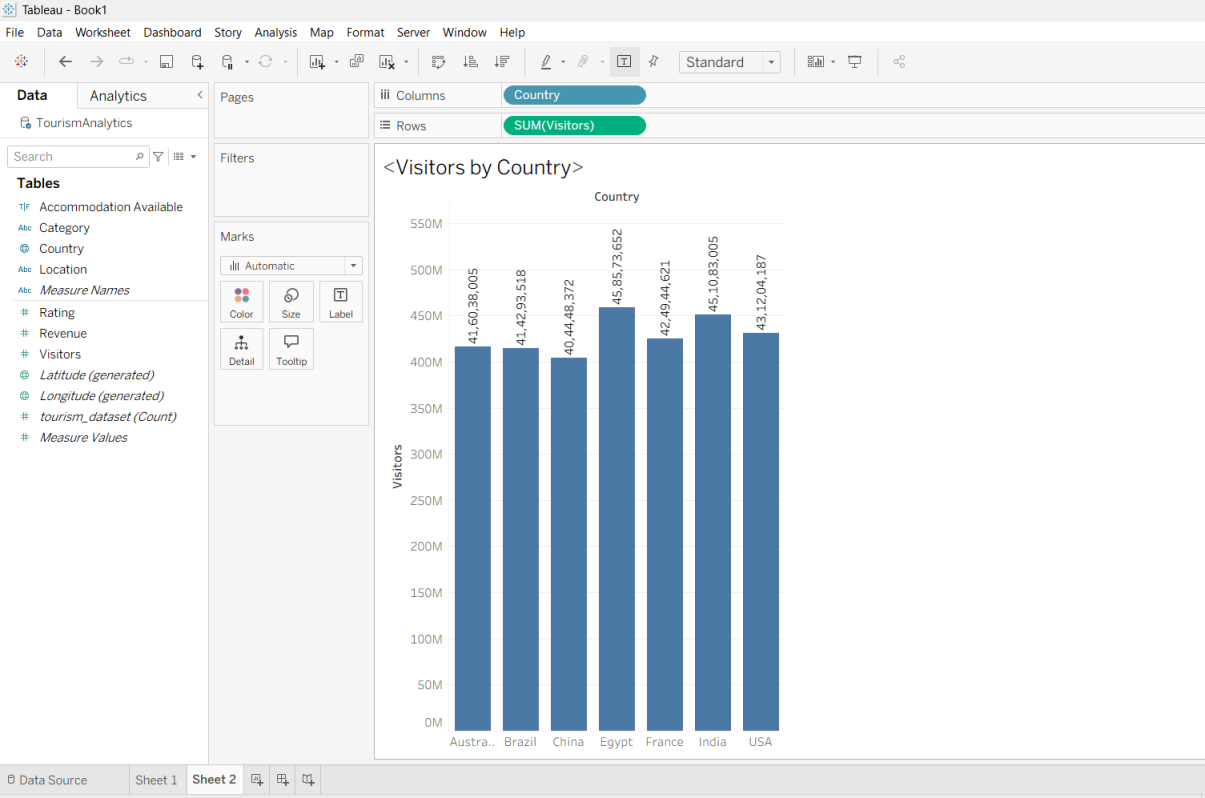


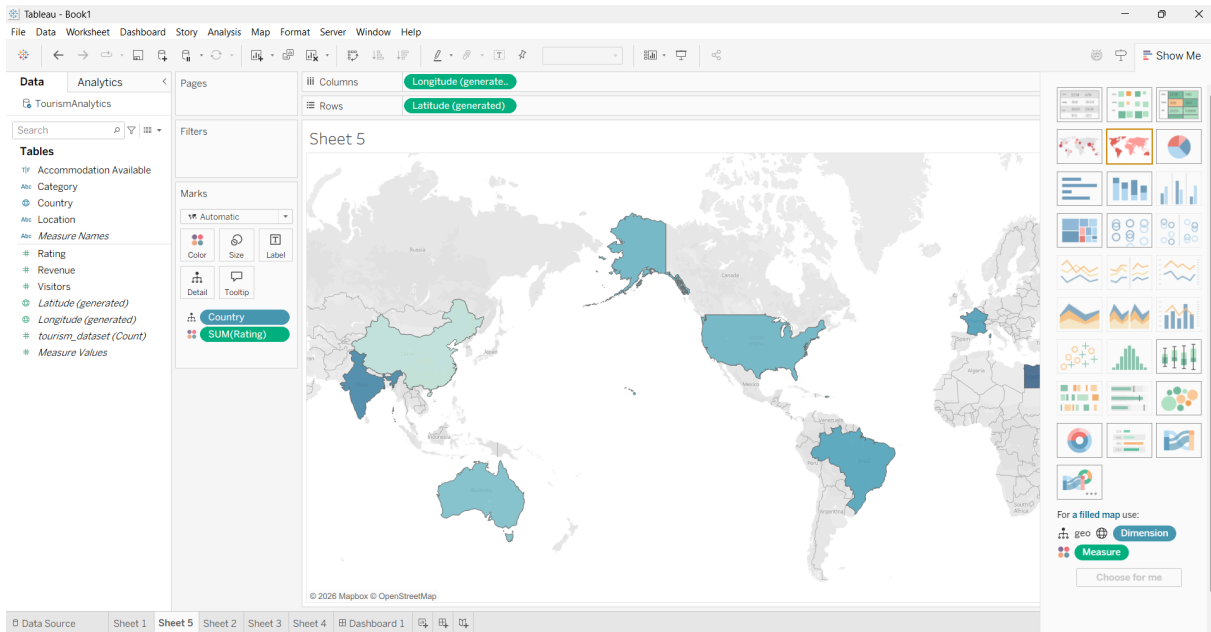
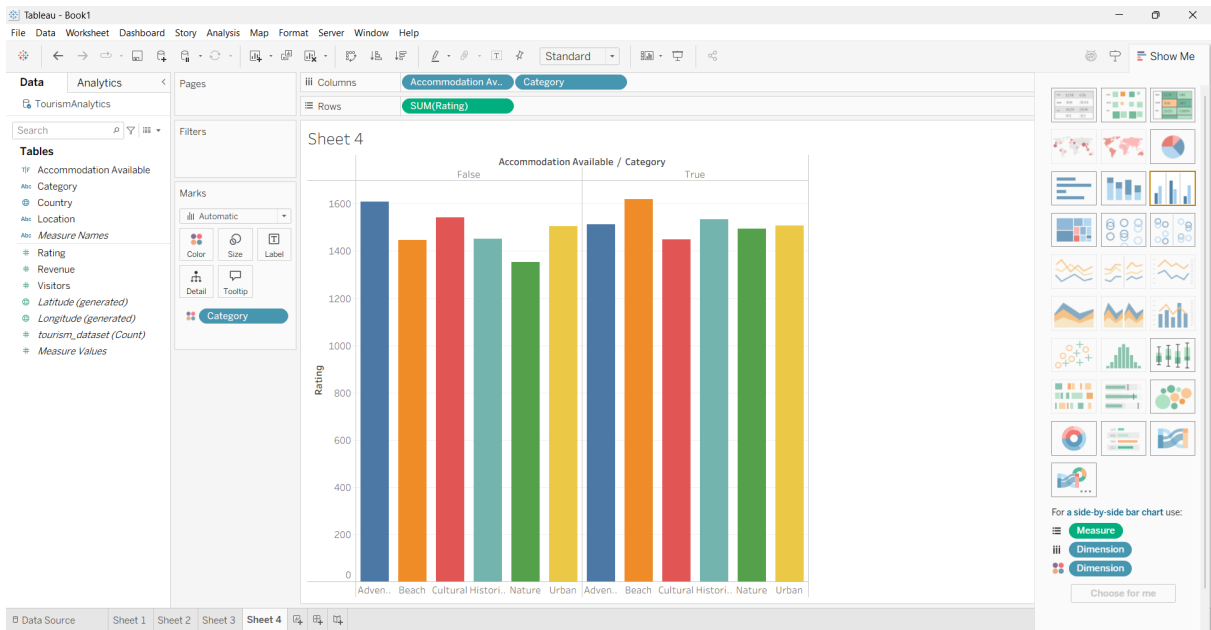


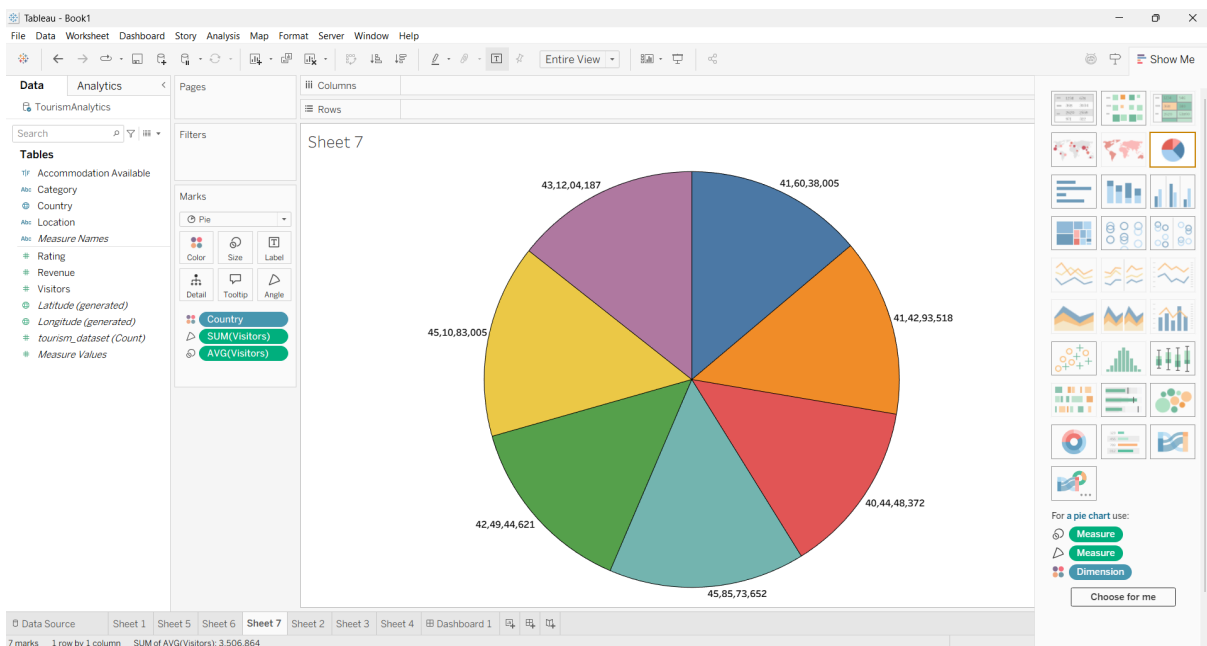
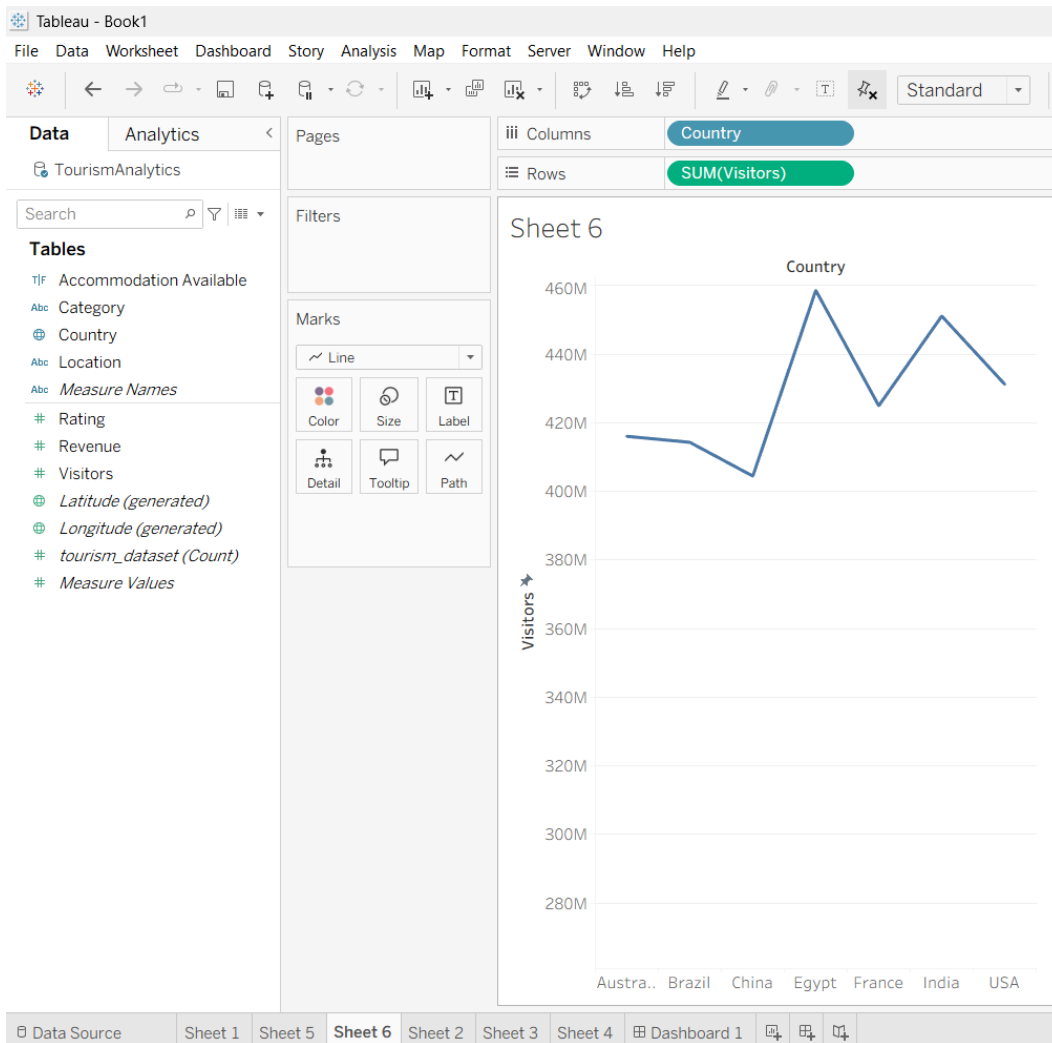
5. Tableau Dashboard

Created visuals in Tableau in separate sheets and created a dashboard

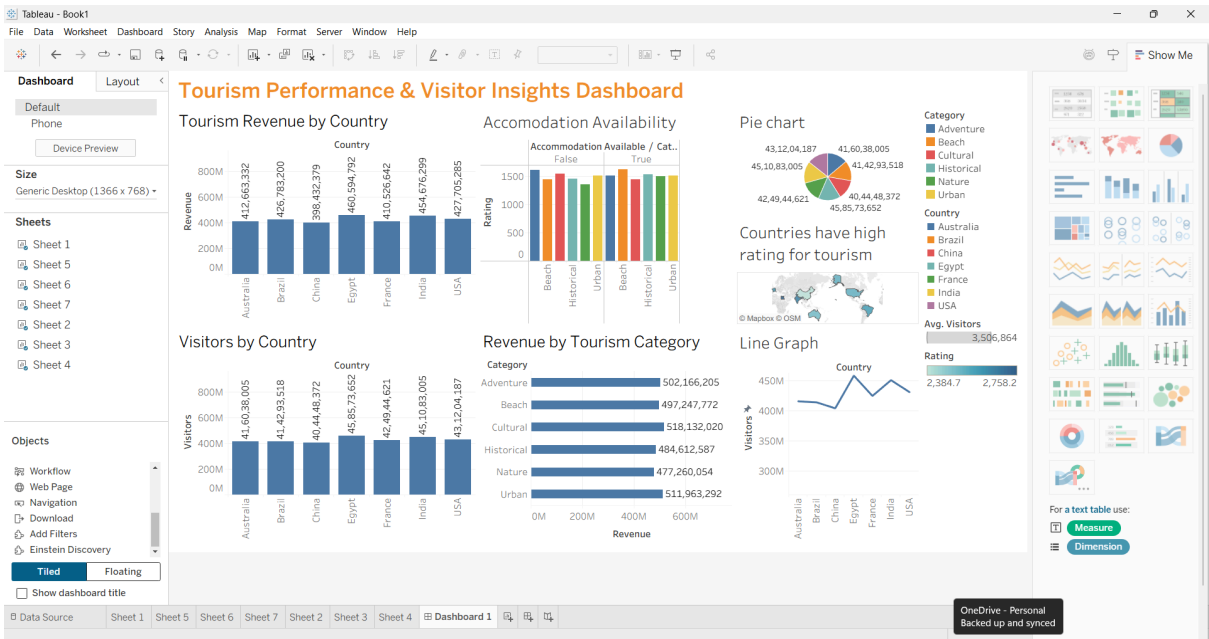








Tourism Performance & Visitor Insights Dashboard



Result: